Restaurant Tip Prediction Using Microsoft Excel

1. Introduction

The aim of this project is to analyze restaurant customer data and develop a predictive model in Microsoft Excel to estimate tip amounts. The dataset includes details about customers, their dining experience, and total bill values. By utilizing Excel's Data Analysis ToolPak, a regression-based model is created to predict tips based on key influencing factors.

2. Dataset Description

The dataset contains the following variables:

- sex: Gender of the customer
- smoker: Indicates whether the customer is a smoker
- day: Day of visit
- time: Lunch or Dinner
- size: Number of people at the table
- total_bill: Total bill amount in USD
- tip: Tip amount in USD (Target Variable)

3. Data Cleaning

The dataset was checked for missing values using Excel (Home → Find & Select → Go to Special → Blanks). No missing values were found.

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is conducted to gain a comprehensive understanding of the dataset, uncover patterns, identify potential outliers, and examine relationships among variables. This process supports effective data cleaning and provides valuable insights for selecting appropriate features and modeling techniques for further analysis.

4.1 Pivot Table: Tip by Gender

Create Pivot Table

1. Click anywhere inside the table.
2. Go to **Insert → PivotTable**
3. Choose: *From Table/Range*
4. Select **New Worksheet**
5. Click **OK**

Set Fields

- Drag **Gender (sex)** → Rows
- Drag **Tip** → Values

Steps to note:

- Click dropdown in Values → **Value Field Settings**
- Choose **Sum**

Now you have **Total Tip by Gender**

Create Pie Chart (Tip by Gender)

1. Click inside Pivot Table.
2. Go to **Insert → Pie Chart → 2-D Pie**
3. Add:
    - Chart Title: *Tip Distribution by Gender*
    - Data Labels (Right click → Add Data Labels)

4.2 Pivot Table: Tip by Smoker

Create Pivot Table

1. Insert new Pivot Table.
2. Drag:
    - **Smoker** → Rows
    - **Tip** → Values (Sum)

Create Pie Chart (Tip by Smoker)

1. Select Pivot Table.
2. Insert → Pie Chart → **3-D Pie**
3. Title: *Tip Distribution by Smoker Status*

4.3 Pivot Table: Time (Lunch/Dinner) by Gender

Create Pivot Table

1. Insert Pivot Table.
2. Drag:
    - **Gender** → Columns
    - **Time (Lunch/Dinner)** → Rows
    - **Tip** → Values (Sum)

Now you see **How much each gender tips during Lunch and Dinner**.

Create 2D Column Chart

1. Click inside Pivot Table.
2. Insert → Column Chart → Stacked Bar
3. Title: *Lunch vs Dinner Tips by Gender*

4.4 Pivot Table: Total Bill vs Tip

Since total bill and tip are numeric variables, you do this differently.

Option A (Recommended for EDA)

1. Insert Pivot Table.
2. Drag:
    - **Total Bill** → Values (Sum)
    - **Tip** → Values (Sum or Average)
3. **Tip%** → Values (Sum)

Steps for creating new Field:

- Click inside Pivot Table.
- PivotTable Analyze →Fields, items, & Sets →Calculated Field →
- Add: Name, Formula
- Click Ok.

Now you see: **Tip amount by bill range.**

4.5 Scatter Chart (Total Bill vs Tip)

Pivot Tables are NOT ideal for scatter.

Create Scatter Plot

1. Select:
    - Total Bill column
    - Tip column
2. Go to **Insert → Scatter → Scatter**
3. **Go chart elements → Trend line →linear**
4. Title: *Total Bill vs Tip*

This shows tipping behavior clearly.

4.6 Scatter Chart (Tip by Size)

Pivot Tables are NOT ideal for scatter.

Create Scatter Plot

1. Select:
    - Size Column
    - Tip Column
2. Go to **Insert → Scatter → Scatter**
3. **Go chart elements → Trend line →linear**
4. Title: *Tip vs Size*

5. Encoding Categorical Variables

Categorical variables were converted into numeric format using IF formulas in Excel:

1. Gender = IF(sex='Male',1,0)
2. smoker = IF(smoker='Yes',1,0)
3. time = IF(time='Dinner',1,0)
4. Day variables were encoded using dummy variables (Fri, Thur, Sat, Sun)
    a. Fri = IF(day=’Fri’,1,0)
    b. Thur = IF(day=’ Thur’,1,0)
    c. Sat = IF(day=’ Sat’,1,0)
    d. Sun = IF(day=’ Sun’,1,0).

6. Model Building (Regression)

Regression was performed using Excel Data Analysis ToolPak. Go to the **Data** tab → click **Data Analysis** → select **Regression** → choose input Y range and nput X range (include labels) → check **Labels in First Row** → choose an output location → click **OK**.

Dependent Variable (Y): tip
Independent Variables (X): encoded categorical variables, table_size, bill_amount.

Regression Equation:

Predicted Tip=b0+b1(sex)+b2(smoker)+b3(Fri)+b4(Thur)+b5(Sat)+b6(Sun)+b7 (time)+b8(size)+b69(total bill)
Where:

- b_0= Intercept
- $b_1$to b_9= Regression coefficients

7. Correlation

Correlation was performed using Excel Data Analysis ToolPak (Go to the **Data** tab → click **Data Analysis** → select **Correlation** → choose your input range (include labels) → check **Labels in First Row** → choose an output location → click **OK**.

The correlation matrix shows relationships between encoded categorical variables, size, total bill, and tip.

Results:
The correlation analysis indicates that both table size and total bill show the strongest positive relationship with tip amounts. Therefore, these two variables were selected as the primary predictors in the regression model due to their greater influence on tipping behavior.

8. Predicted Tips Calculation

Multiple linear regression was used to examine how the size and total bill affected the tip amount prior to estimating tips. The size (number of guests) and total bill are independent variables in this model, whereas tip is the dependent variable.

Compared to a basic linear regression model, our method increases prediction accuracy and enables us to investigate the combined impact of both predictors on tipping behavior. To perform multiple linear regression in Excel, go to the Data tab and click Data Analysis. Select Regression and click OK.

Predicted tips were calculated using the regression equation in Excel for each observation.

Predicted Tip=0.6689+0.1926(size)+0.0927(total_bill)

Where:

- size = number of people at table
- total_bill = bill amount

9. Squared Error Formula Using POWER function

$$\text{Squared Error} = (\text{Actual Tip} - \text{Predicted Tip})^2$$

Excel Formula:

=POWER (Actual_Tip_Cell - Predicted_Tip_Cell, 2).

10. RMSE Calculation

RMSE (Root Mean Square Error) measures how far your predicted values are from the actual values on average. It tells you how much error your regression model is making.

RMSE Formula Structure

RMSE=√((∑(Actual-Predicted) ^2)/n)

In Excel:

=SQRT(SUM(range_of_squared_errors)/COUNT(range_of_squared_errors))

Conclusion

This research effectively illustrated the usage of Microsoft Excel for multiple linear regression predictive analytics. To forecast restaurant tips, a regression-based model was created, the dataset was clean, and categorical variables were successfully encoded. The model's accuracy was assessed using the RMSE measure, which verified that group size and bill amount have a major impact on tip behavior.